Challenge :	The Marauder's Map
Category :	OSINT (Open Source Intelligence) ,Digital Forensics
Difficulty :	Moderate
Tools used :	Chrome browser, Google Maps , Nmap , Exif

Summary

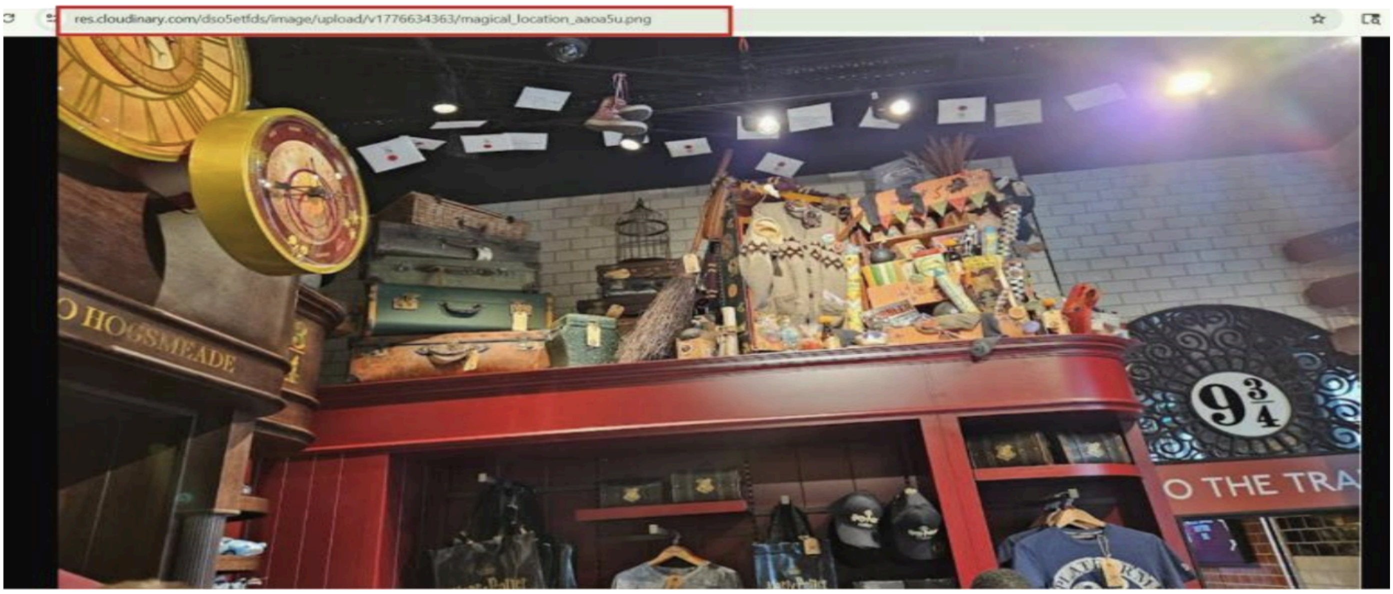
This challenge involves analyzing an image to uncover hidden information not visible at first glance. By extracting the image's metadata using tools like ExifTool, the player finds embedded GPS coordinates. These coordinates are then used in a map service to locate the exact real-world building. The final step is to identify the correct, full name of that location and submit it in the required format

Problem Statement

During a sweep of intercepted communications, the Department of Magical Law Enforcement recovered a single photograph suspected to be a dead drop left by Snape.
The image itself reveals only a facade, yet digital scrying suggests the author embedded precise geographic anchors deep within its invisible structures.
If a tracker were to extract these spatial coordinates and map them to their true physical counterpart, they might uncover the exact building where the fragment is hidden.
However, the Map's verification ward is strict. It demands absolute mathematical precision alongside the location's true, unabbreviated name.
To declare your findings, establish a **n**etwork **c**onduit directly to 80.225.207.209 using portal 9001.
Extract the anchors. Name the destination. Prove that you know exactly where to look.

Steps to solve

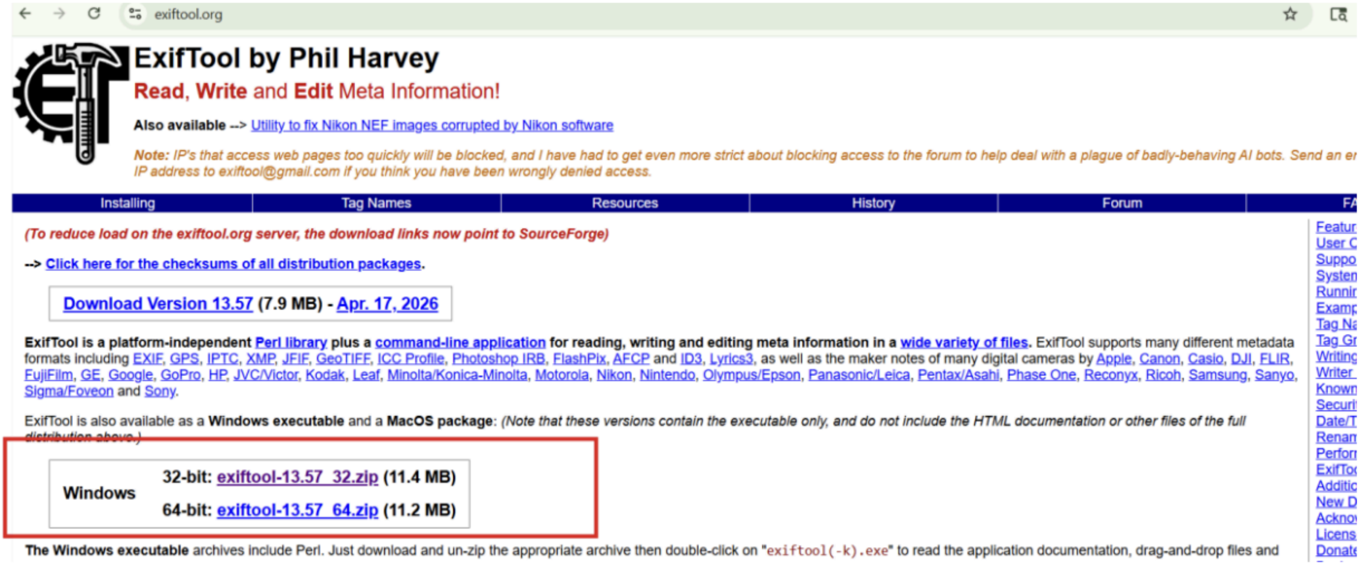
Step 1: Download the image
• Open the given link
• Save the file as magical_location.png on your system



Step 2: Install a metadata tool

• Download and install ExifTool

• This tool helps you read hidden data inside images



Step 3: Extract metadataStep:

• Open terminal/command prompt in the image folder and Run:

cd exiftool-13.57_32

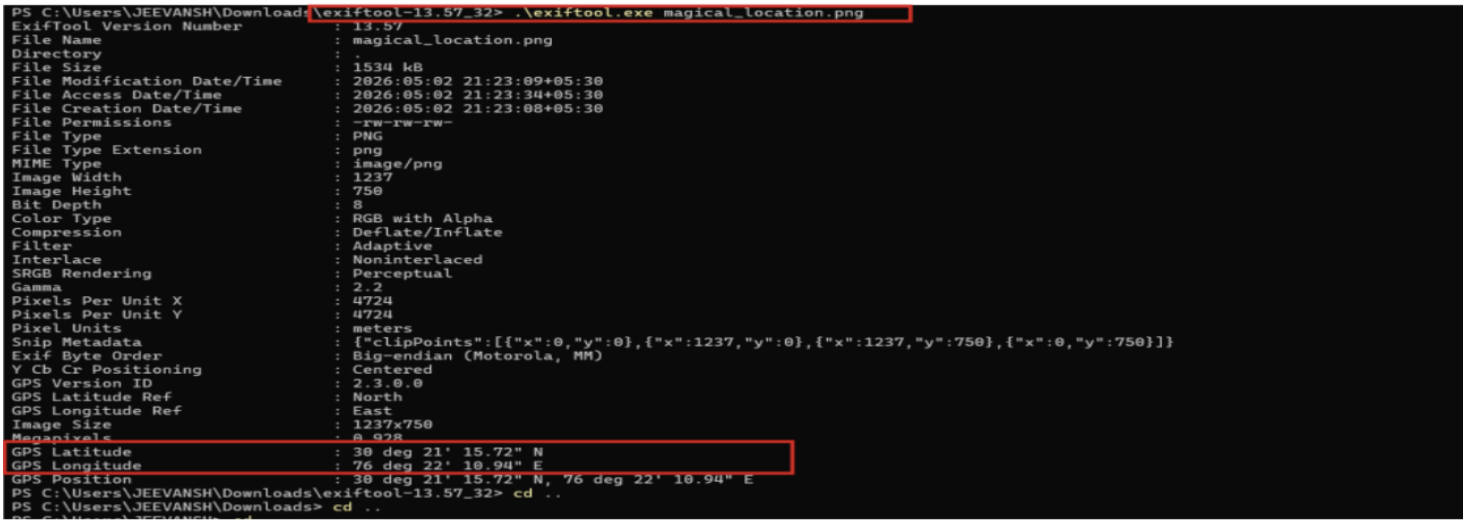
.\exiftool.exe magical_location.png

Step 4: Find GPS coordinates

• Look for:

• GPS Latitude

• GPS Longitude

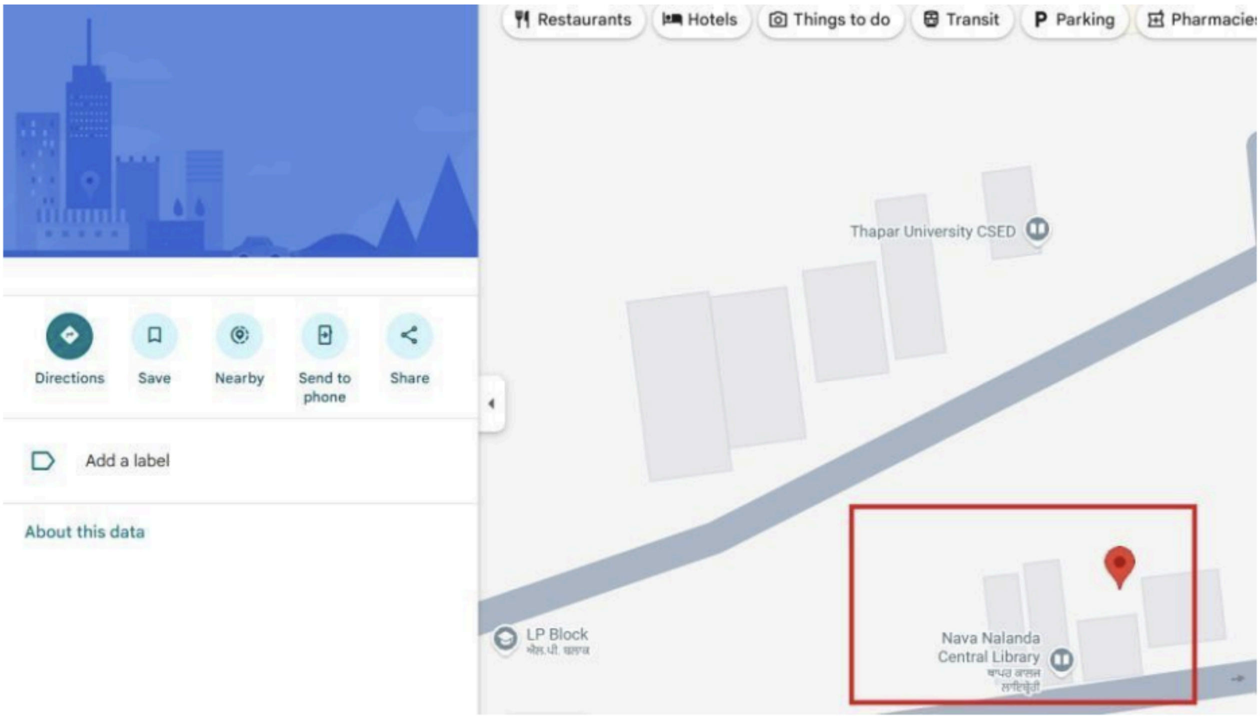


Step 6: Locate on map

• Open Google Maps

• Paste coordinates

• Zoom in and click the building



Step 7: Identify exact location name
- Find the full official name of the place
- HERE : Nava Nalanda central library
- The format of input should be latitude, longitude, place

Step 8: Connect to server
- Use netcat (via Nmap's ncat): ncat 80.225.207.209 9001

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C:\Users\JEEVANSH>"C:\Program Files (x86)\Nmap\ncat.exe" 80.225.207.209 9001

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Step 9: Submit answer in exact format:
30.354368,76.369706,nava nalanda central library

Step 10: Get the flag
CRUXHUNT{m4r4ud3r5_m4p_l34d5_t0_c0ll3g3_l1br4ry}

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The Map does not reveal everything at once...
Your target must be within 100 meters of the true location.

Submit in format: latitude,longitude,location_name
Example: 30.1234,76.5678,Some Place

> 30.354368,76.369706,nava nalanda central library

Γ£à CORRECT! You are 0 meters from the target.

The Map shifts... the hidden path reveals itself.
≡fôì Location Verified: Nava Nalanda Central Library

≡fÄë Here is your flag:
CRUXHUNT{m4r4ud3r5_m4p_l34d5_t0_c0ll3g3_l1br4ry}

Mischief managed.
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